

Dear examiner,

please find the criteria marks as well as teacher's comments below.

A: 4	B: 3	C: 3	D: 2	E: 4	Total: 16
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Criterion A: Presentation

A very well-written and easy to follow IA. The introduction has a clear rational and aim, which is fully met. The exploration's structure is well-developed (introduction, body, conclusion); all sections link well to each other and the reader never feels lost. Equations and a number of useful diagrams and pictures are used effectively to aid understanding. There are no irrelevant or repetitive calculations.

Criterion B: Mathematical communication

The student uses relevant, appropriate mathematical communication. Equations, diagrams and tables are used effectively. All key terms and variables are defined.

Criterion C: Personal engagement

The IA shows outstanding personal engagement. The student took a creative approach investigating a very interesting real-life situation of his choice. The IA contains almost exclusively unfamiliar mathematics that is all self-taught and explained in a very unique way such that a fellow student can comprehend. The student also taught himself how to work with the 3D engine within the Math software Geogebra and found a creative and comprehensive way of checking if his answer is reasonable by estimating the chip's area using paper as well as crushing a chip and laying the pieces out on square paper.

Criterion D: Reflection

Meaningful reflection is seen throughout the IA (the student successfully keeps linking back to his exploration's aims) as well as at the end where the student summarizes his findings, discusses limitations to his modeling, and states two possible extensions to his work. However, most of the reflection seen is not critical.

Criterion E: Use of mathematics

The student uses relevant mathematics commensurate with the level of the course. All mathematical arguments are clear and logical; all mathematics explored is error-free and an appropriate degree of accuracy is used at all times. Good knowledge and understanding of a number of unfamiliar concepts are demonstrated - the student successfully applies new mathematics and "plays" with it. However, the student mostly doesn't demonstrate sophistication or rigour, which he could have done by, for example, sketching an approach to solve the definite integral on page 17.

Modeling and Calculating the Surface Area of a Pringles Chip

hdv591

Mathematics: Analysis and Approaches Higher Level
Internal Assessment

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1. Introduction

1.1 Rationale

Potato chips have varieties of flavors made by various brands and they come in different size and shapes. Pringles is one of the popular American chips brand. It is packaged in the iconic cylindrical can with the Mr. Pringles logo. The saltiness as well as the crispy and crunchy texture makes the consumers addicted to Pringles. I am one the consumers who is addicted to Pringles and enjoy eating it. Whenever I eat Pringles, I always question why the chips have their particular and unusual curved shape. I decided to research to figure out the reasons behind the shape. The answers I found were:

- It's easier to stack the chips in the cylindrical can (Bhattacharya).
- Minimizes the risk of the chips to be broken when it is transported (Bhattacharya).
- Despite the thin shape, it gives structural strength (Villaluz).

Additionally, I found out the shape was developed by Fredric J. Baur who was a chemist and also was an expert in mathematics (Hitt). The shape has a name called “saddle” or in mathematical terms it is called a “hyperbolic paraboloid” (Bhattacharya).



Figure 1: Image of Pringles chips (Glane23)

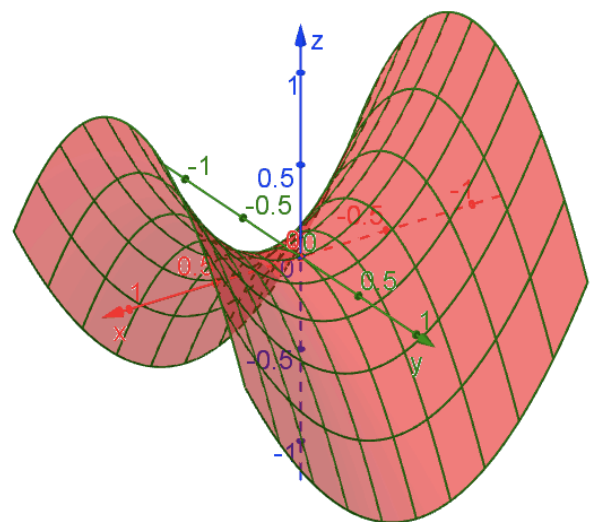


Figure 2: Image of hyperbolic paraboloid (Lo)

Through further research of hyperbolic paraboloid, I found connections to the Mathematics AA HL syllabus such as relations to vectors and calculus. In the syllabus, we learned about 2 dimensions and also 3 dimensions but we only looked at planes. I considered that an investigation on the hyperbolic paraboloid could develop my understanding of the vectors unit by working with curves and complicated shapes. Moreover, the surface area of a hyperbolic paraboloid could be calculated using surface integrals; specifically using double integrals. This could also develop my understanding on the calculus unit. For these reasons, I decided to do an investigation on a Pringles chip.

1.2 Aims

The aim of this exploration is to model a Pringles chip and to find the surface area of one side. In order to do so, first the characteristics and equation of a hyperbolic paraboloid need to be explored and understood. Then using the equation, a Pringles chip will be modeled using the math software GeoGebra. Before calculating the surface area, different methods will be used to estimate the surface area of a Pringles chip. Afterward, the surface area will be calculated based on the model. To calculate the surface area, double integrals will be used.

2. Modeling a Pringles Chip

2.1 Equation and Characteristics of Hyperbolic Paraboloids

Quadric surfaces are three-dimensional extensions of parabolas, hyperbolas, and ellipses which are also known as conics (J). From Mathematics AA HL syllabus, we know that the cartesian equation of the plane is $ax + by + cz = d$ which is a first-degree equation with 1 as the exponent of x , y , and z . The equation for quadric surface will always include x^2 , y^2 , and/or z^2 meaning it will always be a second-degree equation with 2 as the exponent of x ,

y , and z (OpenStax College). Hyperbolic paraboloid is one type of quadric surface. The general formula for hyperbolic paraboloid is:

$$\frac{z}{c^2} = \frac{x^2}{a^2} - \frac{y^2}{b^2} \quad (1)$$

Where a , b , and c are constants (J).

There are different types of quadric surfaces and to identify whether it is a hyperbolic paraboloid or not, the equation must be examined. An equation containing x^2 , y^2 , z , and a negative sign, which is shown in Equation (1), will be a hyperbolic paraboloid (Mathispower4u).

A hyperbolic paraboloid has two parabolas in the vertical cross-section and one hyperbola in the horizontal cross-section. The name hyperbolic paraboloid originates from the fact that it has a hyperbola. In every case, one parabola is concave down and another is concave up. From Figure 3, it can be seen that the blue curves are concave down parabolas, the red curve is a concave up parabola, and the green curves are hyperbolas. The hyperbola opens up in either the direction of x -axis or y -axis depending on the value of z in Equation (1) (Rogness).

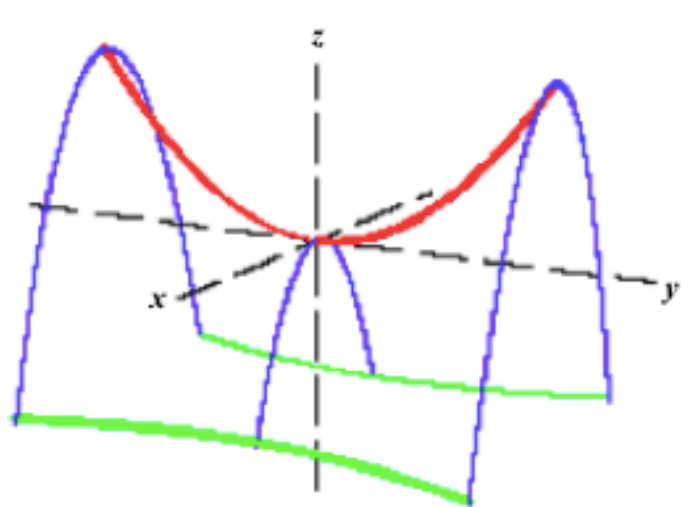


Figure 3: Graph of a hyperbolic paraboloid indicating two parabolas and a hyperbola (Encyclopedia of Mathematics)

By using Equation (1), it can be shown why a hyperbolic paraboloid has two parabolas and one hyperbola. When considering a , b , and c to be equal to 1 and $x = 0$, the equation will be:

$$z = -y^2 \quad (2)$$

As there is a negative sign and has a squared term, we understand the equation gives a concave down parabola. Now when considering a , b , and c equal to 1 and $y = 0$, the equation will be:

$$z = x^2 \quad (3)$$

As it is positive and has a squared term, we understand the equation provides a concave up parabola. To visualize Equation (2) and (3), I used GeoGebra. The results can be seen in Figure 4.

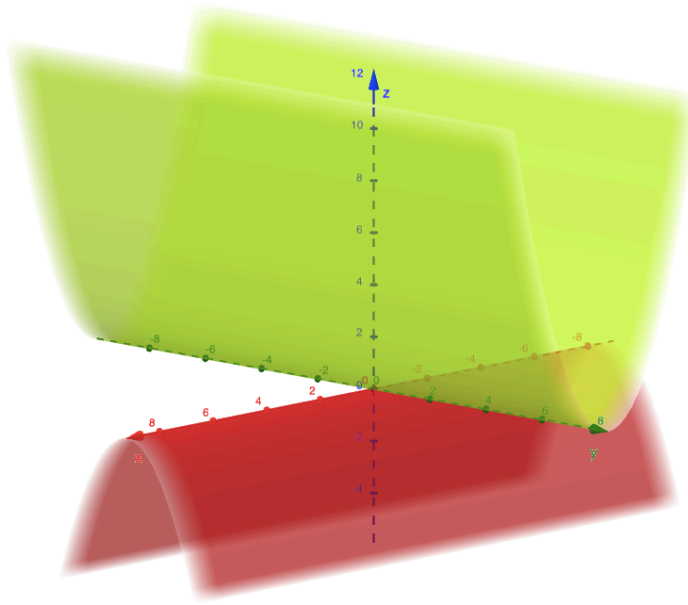


Figure 4: Green shows $z=x^2$ and red shows $z=-y^2$ (GeoGebra)

A hyperbola can also be found using Equation (1), where $z = 0$ and when a , b , and c are equal to 1. Then we have the equation:

$$x^2 - y^2 = 0 \quad (4)$$

When $z = 0$, it will simply provide two planes intersecting at z -axis which can be seen on Figure 5 on the next page. However, when the value of z is either $z > 0$ or $z < 0$, it will give a hyperbola. When $z > 0$, the hyperbola opens up in the direction of x -axis and when $z < 0$, the hyperbola opens up in the direction of y -axis. I used two equation with a value of z that meets the requirements. The two equations are:

$$x^2 - y^2 = 10 \quad (5)$$

$$x^2 - y^2 = -10 \quad (6)$$

The graph of Equation (5) and (6) can be seen on Figure 6.

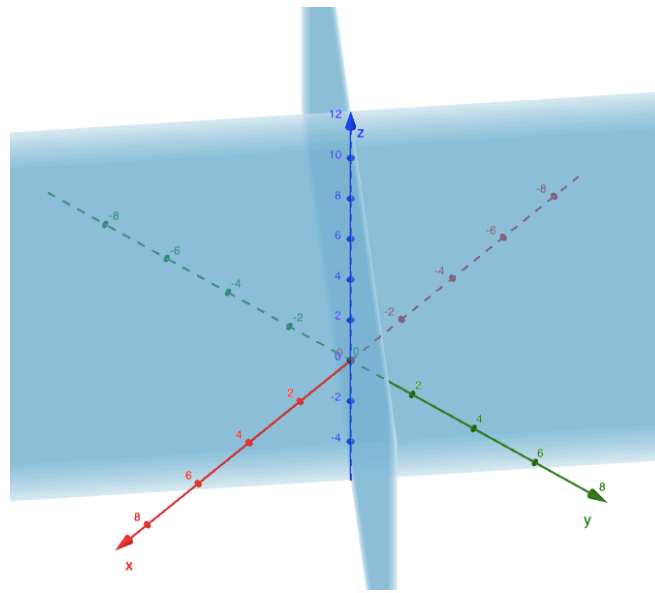


Figure 5: Graph of $x^2 - y^2 = 0$ (GeoGebra)

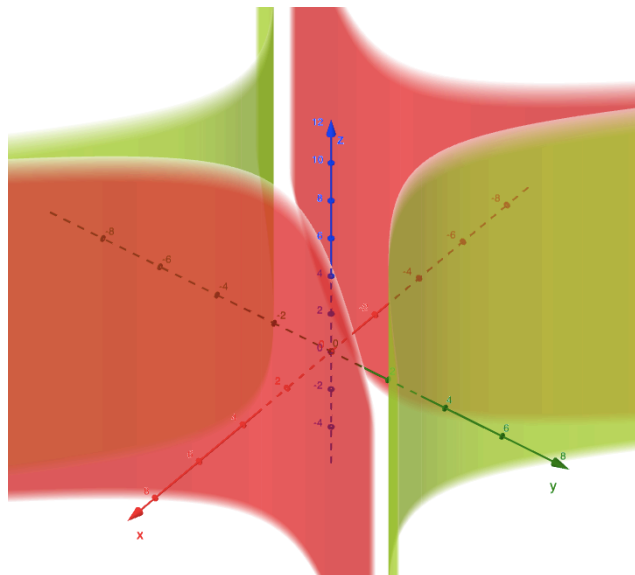


Figure 6: Graph of $x^2 - y^2 = 10$ (Red) and $x^2 - y^2 = -10$ (Green) (GeoGebra)

A hyperbolic paraboloid has double curve and this can be identified by looking at Figure 7. It can be seen that the curve on the x-axis and y-axis plane, which is the direction of the movement of the two parabolas, create the double curve. The green parabola moving in x-axis plane is one curve and the purple parabola moving in y-axis plane is another curve. At the origin $(0, 0, 0)$, the maximum point of one curve meets the minimum point of the other curve. This point is known as the saddle point or a minimax point (Villaluz). In the case of Figure 7, the saddle point is where the maximum point of the curve on the y-axis plane and

minimum point of the curve on the x-axis plane meet. Additionally, hyperbolic paraboloids are non-developable, meaning the curve cannot be flattened as a sheet.

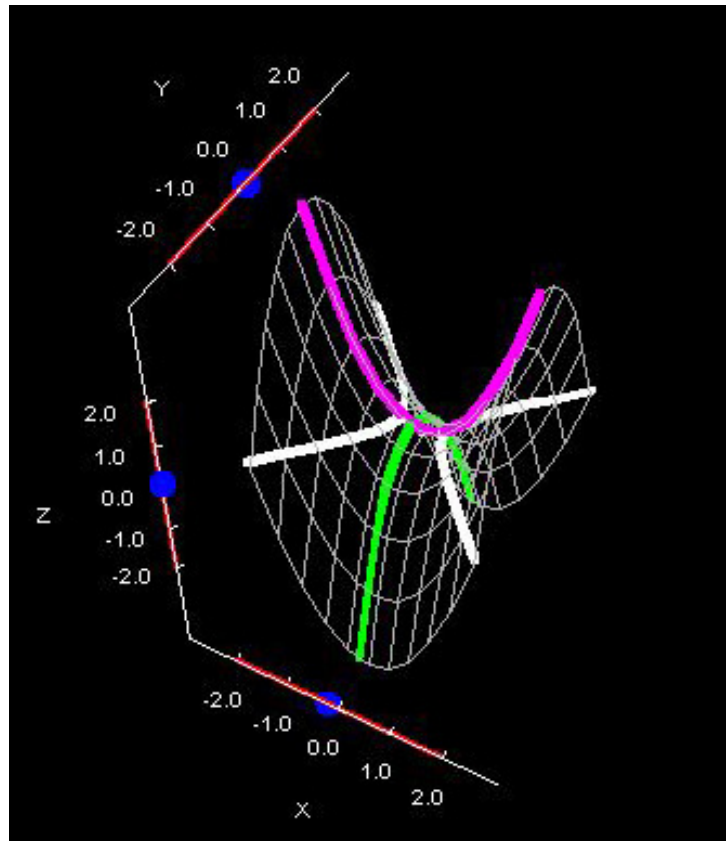


Figure 7: An image showing the double curve of a hyperbolic paraboloid (Kerning)

2.2 Graphing a Pringles Chip

In order to graph a Pringles chip, GeoGebra will be used. Before graphing it, I used a ruler to measure the different parts of a Pringles chip. The distance between the two minimum points of a curve was measured to determine the length. For determining the width, the two maximum points of a curve was measured. Looking at Figure 8, it can be seen where the two maximum and minimum points of the curve on a Pringles chip are. Figure 9 and 10 demonstrates how the length and width were measured using a ruler. The length and width of a Pringles chip is indicated on Figure 11. The length was 6.2cm and the width was 4.5cm. The height of a Pringles chip was also measured by placing it on a flat surface. Then the distance between the surface and the maximum point was measured. By identifying the

dimensions, we can estimate the points on a Pringles chip. This will help to estimate the equation of a hyperbolic paraboloid to create an accurate model which goes through all the points.

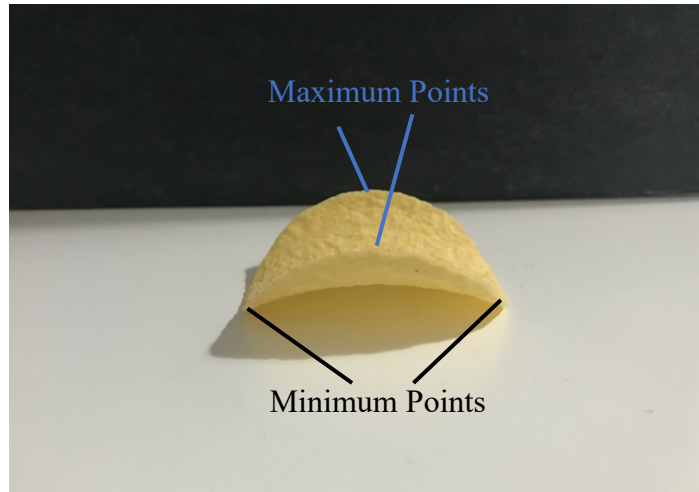


Figure 8: Indicating the maximum and minimum points on a Pringles chip (Self-taken)



Figure 9: Demonstrating how the length of a Pringles chip was measured (Self-taken)

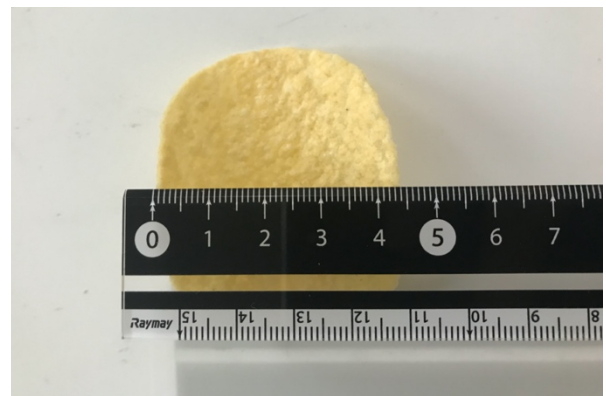


Figure 10: Demonstrating how the width of a Pringles chip was measured (Self-taken)

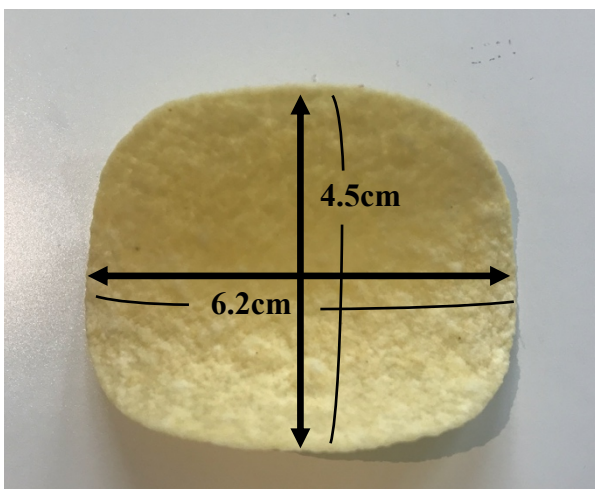


Figure 11: Length and width of a Pringles chip (Self-taken)

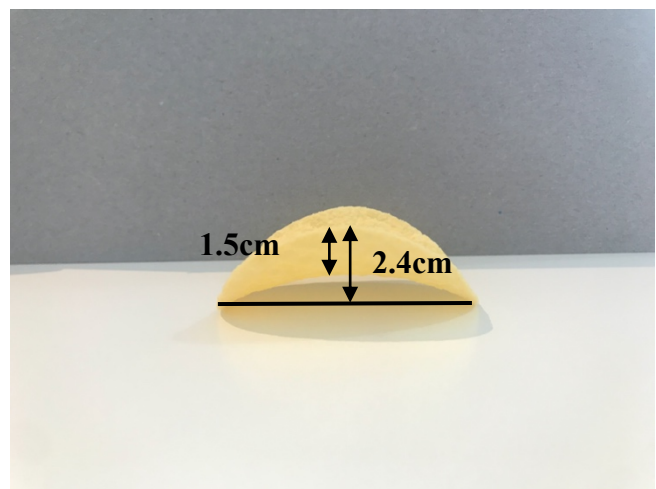


Figure 12: Height of a Pringles chip (Self-taken)

We know from the background research of hyperbolic paraboloid that it goes through the origin $(0, 0, 0)$. The first thing we can do to find other points is to separate the height into above and below the origin. Above the origin is the 1.5cm part on Figure 12. This 1.5cm can be found by measuring the height below the origin using a ruler and it was approximately 0.9cm. Since the height is 2.4cm, subtraction can be used:

$$2.4cm - 0.9cm = 1.5cm \quad (7)$$

The coordinate system is filled as follows: the x-axis will be the width (4.5cm), the y-axis will be the length (6.2cm), and the z-axis will be the height (2.4cm). Considering these factors, we can find the following points.

$$(0, 0, 0), (2.25, 0, 1.5), (-2.25, 0, 1.5), (0, 3.1, -0.9), (0, -3.1, -0.9)$$

The five points express the origin, two maximum, and two minimum points. Only these five points are being used because they are the most accurate points that can be obtained from the measurements.

Now inserting the points in GeoGebra, it will give a visualization of which points the hyperbolic paraboloid needs to go through.

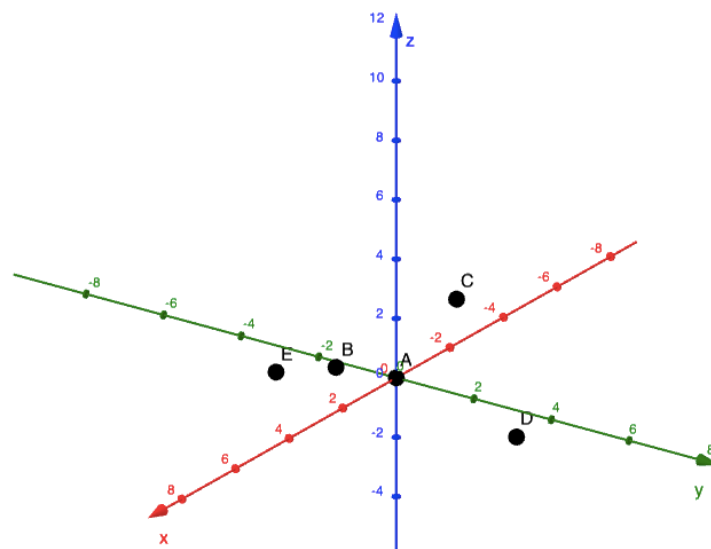


Figure 13: Five points of a Pringles chip (GeoGebra)

Where:

$$A(0, 0, 0), B(2.25, 0, 1.5), C(-2.25, 0, 1.5), D(0, 3.1, -0.9), E(0, -3.1, -0.9)$$

Now we need to find an appropriate equation using Equation (1). The constant b must be larger than 0 meaning $b > 0$. Otherwise, if b is less than 0, sign change will occur, from negative to positive in the equation, resulting in an equation for a different shape. By trial and error, as determining the values of a and b algebraically is outside the scope of this exploration, the best fit equation was found.

$$z = \frac{x^2}{3.5} - \frac{y^2}{11} \quad (8)$$

Figure 14 represents Equation (8) on GeoGebra. The range and domain were restricted to $-2.25 \leq x \leq 2.25$ and $-3.1 \leq y \leq 3.1$ respectively.

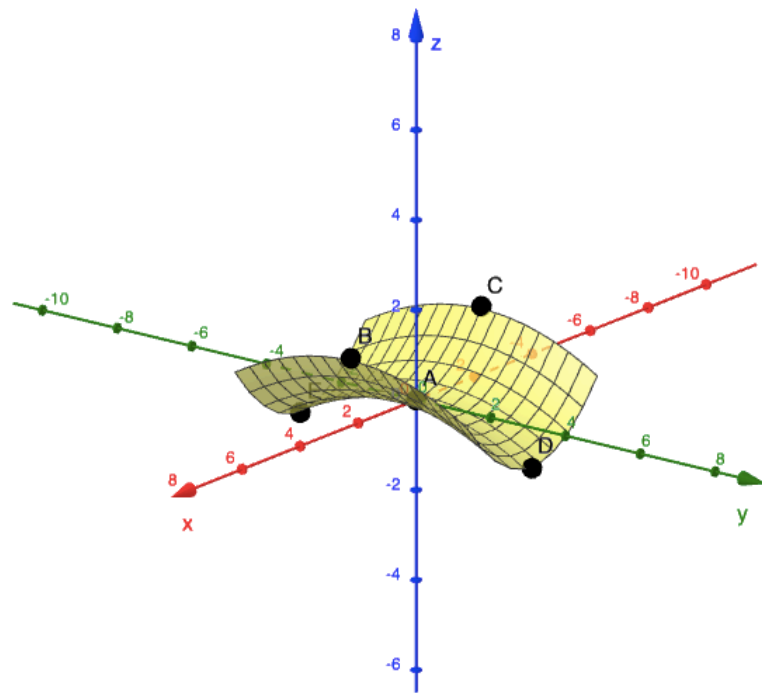


Figure 14: Graph of Equation (8) showing that it passes through the five points (GeoGebra)

The shape of a Pringles chip is starting to appear but with the current equation it does not provide the roundness. Therefore, I would like to introduce a new formula which will help to present the roundness. The equation is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (9)$$

where a and b are constants (Ferreol).

This is the equation for an elliptic cylinder. Like its name, this shape is a cylinder with ellipse on the top and bottom (Ferreol). Using Equation (9), an elliptic cylinder which goes through four points, A , B , C , and D , will be found. The constants a and b are the magnitude from the origin to either Point B or C and Point D or E respectively. If Point B and D are taken as the value of the constants the equation will be:

$$\frac{x^2}{2.25^2} + \frac{y^2}{3.1^2} = 1 \quad (10)$$

Graphing Equation (10) in GeoGebra, it will give an elliptic cylinder seen in Figure 15.

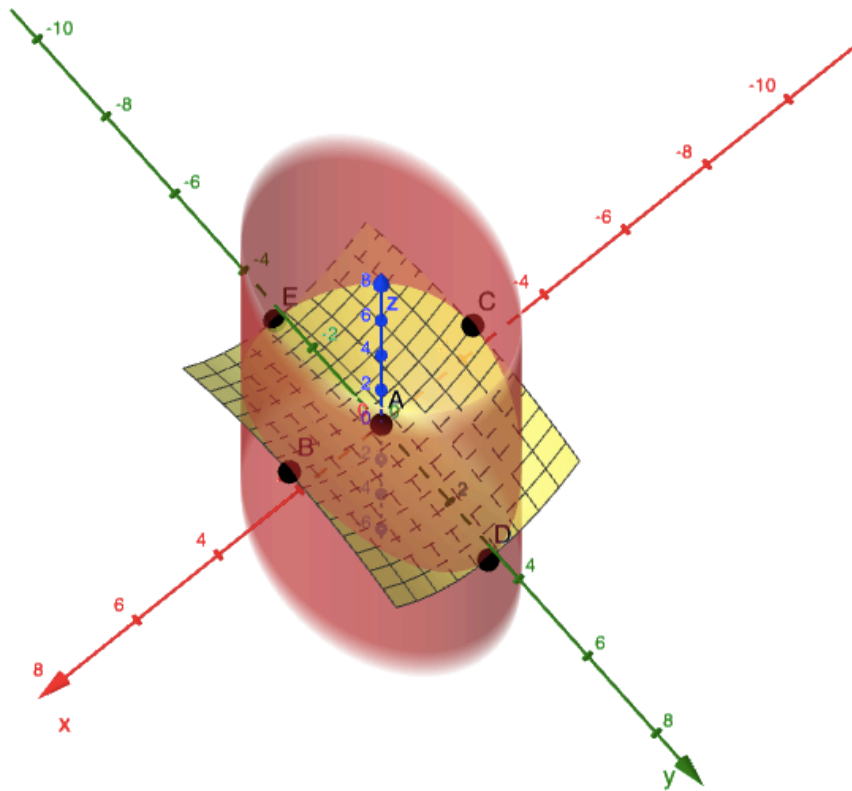


Figure 15: Equation (10) graphed on GeoGebra giving elliptic cylinder going through four points (GeoGebra)

Finally, the model of a Pringles chip is complete after finding the curve of the intersection and filling the inside of elliptic cylinder with the hyperbolic paraboloid found in previous step.

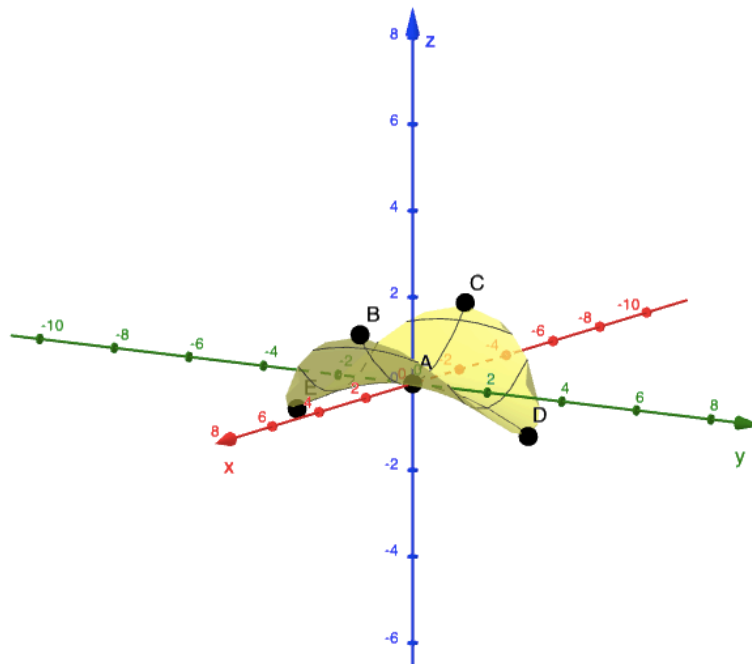


Figure 16: Model of a Pringles chip (GeoGebra)

Although the curvature around the edge of a Pringle chip may not be an exact ellipse as the real chips, it can be considered fairly accurate since it passes through the points required. There should not be a large difference between the real and modeled Pringles. The difference can be considered negligible which would not affect the calculations when finding the surface area.

3. Finding the Surface Area of a Pringles Chip

3.1 Estimating the Surface Area of a Pringles Chip

Before calculating the surface area of a Pringles chip, we can use two different methods to estimate the surface area. The estimations found can be compared with the answer derived by the calculation to determine if the solution is reasonable.

The first method is done using a sticky note. A sticky note will be overlaid on top of a Pringles chip and using scissors the paper will be cut. After cutting the paper out, the diameter of the short and long side can be measured using a ruler. Then the formula can be used to calculate the area of an ellipse. The formula is:

$$Area = \pi \times a \times b \quad (11)$$

Where a is half of the major axis and b is half of the minor axis (Web Formulas).

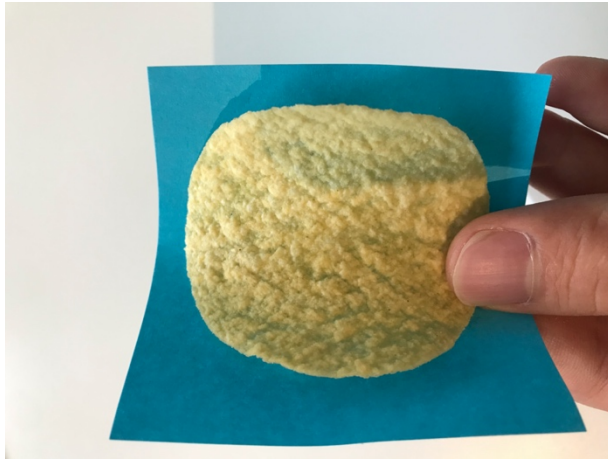


Figure 17: Overlaying a sticky note onto a Pringles chip (Self-taken)

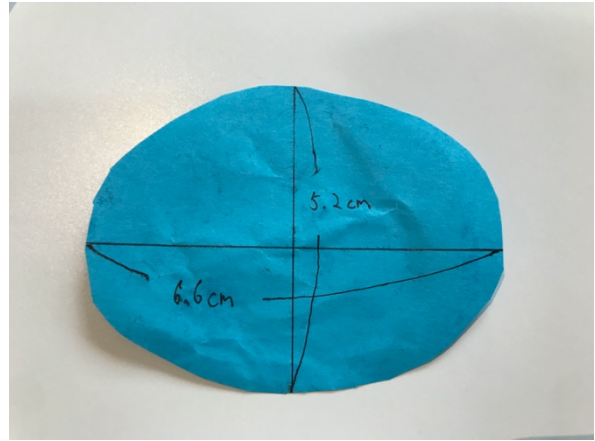


Figure 18: Measuring the diameter of the longest and shortest side (Self-taken)

The major axis is the longer diameter and the minor axis is the shorter diameter of the ellipse. In this case, the major axis will be 6.6cm and the minor axis will be 5.2cm. Since a and b are half of the axes, the two values need to be divided by two.

$$a = 6.6 \div 2 = 3.3 \quad (12)$$

$$b = 5.2 \div 2 = 2.6 \quad (13)$$

Now that a and b are obtained in Equation (12) and (13), it can be substituted into Equation (11) to find the area of an ellipse.

$$Area = \pi \times 3.3 \times 2.6$$

$$Area = 26.95486497 \approx 27.0 \quad (14)$$

The answer is rounded to three significant figures, giving the approximated surface area as 27.0cm^2 . However, the sticky note was not perfectly overlaid on the Pringles chip, especially

at the curves as it was unable to overlay along. Therefore, we can assume the area to be larger than the value derived in Equation (14).

The second method is done by crushing a Pringles chip on a graph paper. Each square has length of 0.5cm and area of 0.25cm². Placing a Pringles chip on the paper, it will be crushed into pieces by hand as seen on Figure 19. After it is crushed, a square will be drawn around the pieces and the surface area of the square will give an approximate value seen in Figure 20 and 21 respectively. To find the area of the square, the length of the two sides need to be multiplied:

$$7 \times 6 = 42 \quad (15)$$



Figure 19: Crushing a Pringles chip on a graph paper (Self-taken)

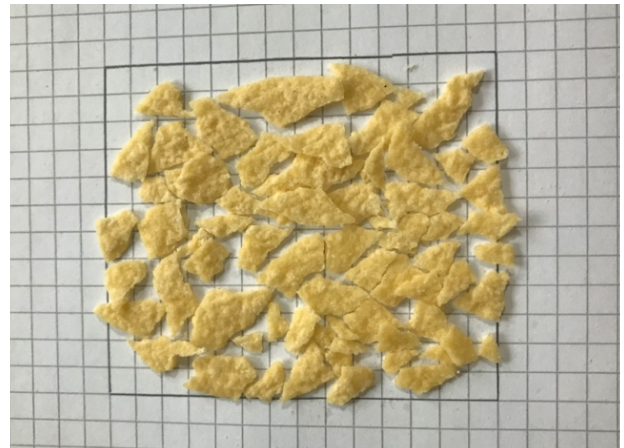


Figure 20: Drawing a square around the crushed Pringles chip (Self-taken)

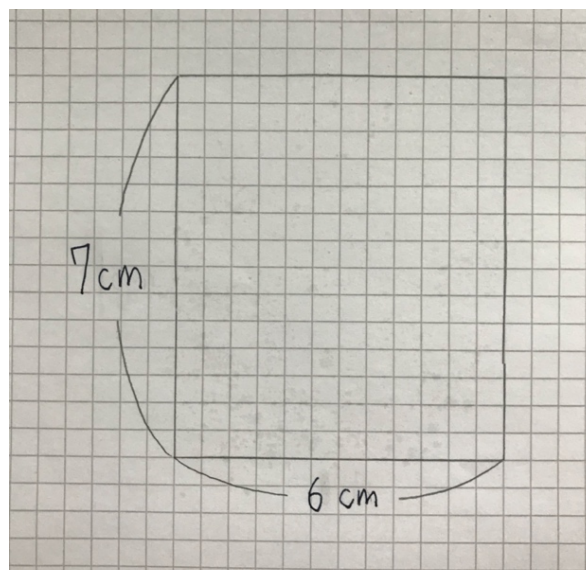


Figure 21: The dimension of the square (Self-taken)

The area of the square is 42cm^2 but as seen on Figure 20, there are a lot of space leftover between the broken chips resulting in few empty squares. Thus, the area will be smaller than 42cm^2 . There are approximately 20 empty squares which can be found by counting it on Figure 20. If the square is not possessed by the broken piece by more than half, it is considered as an empty square. Four squares will create 1cm^2 , so by dividing we can see how much cm^2 is empty space.

$$20 \div 4 = 5 \quad (16)$$

By subtracting the answer of Equation (16) from 42cm^2 , the estimated surface area will be 37cm^2 .

The estimate value of the surface area we obtained from the two methods area 27cm^2 and 37cm^2 . Using the two answers, the answer from the calculation should be between 27cm^2 and 37cm^2 on one side.

3.2 Calculating the Surface Area of a Pringles Chip

As mentioned earlier, hyperbolic paraboloids cannot be developed. If the shape was developable, calculating the surface area would have been an easy task. Since it is non-developable, we need to apply multivariable calculus in order to calculate the surface area.

Before introducing the formula to calculate the surface area of a hyperbolic paraboloid, we need to understand the concept of multivariable calculus. The calculus we studied in Mathematics AA HL is known as single variable calculus. We worked with functions with single variable where y is a function of x , $y = f(x)$ (Wyzant). An example of single variable function would be $f(x) = x^2$. In terms of the dimensions, single variable calculus is considered in two-dimension. Multivariable calculus, like its name “multi” suggests, deals with two variables in a function where z is a function of x and y , $z = f(x, y)$

(Wyzant). For example, $f(x, y) = x^2 - y^2$ is a function with two variables. Of course, multivariable calculus is used in three-dimensional space.

The type of multivariable calculus used is called the surface integral where we will need to integrate twice. It is also known as a double integral. The formula that is going to be used to calculate the surface area is:

$$A(S) = \iint_D \sqrt{1 + \left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2} dydx \quad (17)$$

S represents the surface in the three-dimensional space which is the hyperbolic paraboloid of a Pringles chip, thus $A(S)$ means the area of the surface. D is the shape projected on the xy -plane. In Equation (17), D represents that it is integrated within the projected shape D . In this case, D should be an ellipse with domain of $-2.25 \leq x \leq 2.25$ and range of

$-3.1 \leq y \leq 3.1$. The ellipse will be the base of elliptic cylinder on Figure 15. $\frac{dz}{dx}$ and $\frac{dz}{dy}$ are partial derivatives of z with respect to x and y from the equation of the hyperbolic paraboloid which is Equation (8). D on the xy -plane will be integrated twice to get to $A(S)$, the area of a Pringles chip. Figure 22 represents how the surface in three-dimensional space is projected on xy -plane as D . It is just an example; thus, it is not the actual shape we are dealing with.

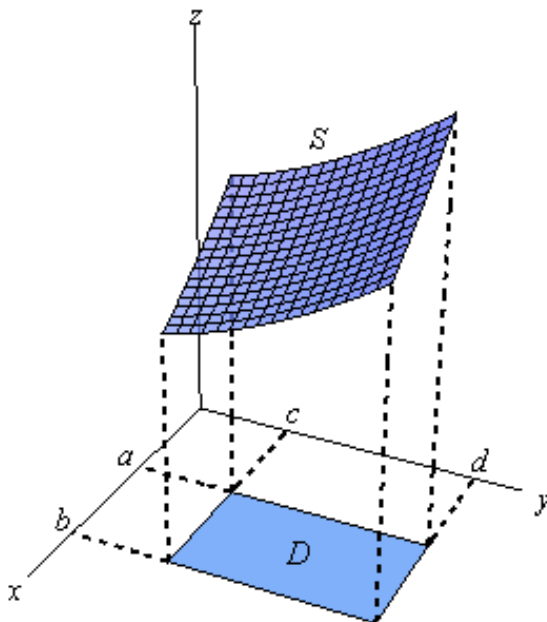


Figure 22: A visual representation of S and D in Equation (17) (Dawkins)

To begin with, partial derivatives need to be calculated. The derivatives of $\frac{x^2}{3.5}$ and $-\frac{y^2}{11}$ from Equation (8) are required. Differentiating these two, it will give:

For $\frac{x^2}{3.5}$:

$$\begin{aligned}\frac{dz}{dx}\left(\frac{x^2}{3.5}\right) &= \frac{dz}{dx}\left(\frac{2}{7}x^2\right) \\ &= \frac{2}{7} \times \frac{dz}{dx}(x^2) \\ &= \frac{2}{7} \times 2x \\ &= \frac{4}{7}x\end{aligned}\tag{18}$$

For $-\frac{y^2}{11}$:

$$\begin{aligned}\frac{dz}{dy}\left(-\frac{y^2}{11}\right) &= -\frac{1}{11} \times \frac{dz}{dy}(y^2) \\ &= -\frac{1}{11} \times 2y \\ &= -\frac{2}{11}y\end{aligned}\tag{19}$$

Now the answers of the derivatives can be substituted into Equation (17):

$$A(S) = \iint_D \sqrt{1 + \left(\frac{4}{7}x\right)^2 + \left(-\frac{2}{11}y\right)^2} dydx\tag{20}$$

Then expanding the brackets:

$$A(S) = \iint_D \sqrt{1 + \frac{16}{49}x^2 + \frac{4}{121}y^2} dydx\tag{21}$$

To solve double integrals, it should be solved from inside to outside, therefore first the integral of y and then integral of x should be calculated. Before calculating the integral, the limits of x and y need to be found. From the dimensions of a Pringles chip, the domain of x

will be $-2.25 \leq x \leq 2.25$ and the range of y will be $-3.1 \leq y \leq 3.1$. By substituting the limits in Equation (21), it gives:

$$A(S) = \int_{-2.25}^{2.25} \int_{-3.1}^{3.1} \sqrt{1 + \frac{16}{49}x^2 + \frac{4}{121}y^2} dy dx \quad (22)$$

A numerical integration software, Wolfram was used to calculate Equation (22) which gives (WolframAlpha):

$$A(S) = 35.5177 \quad (23)$$

Rounding the answer to three significant figures gives:

$$A(S) = 35.5 \quad (24)$$

The approximate surface area of one side of a Pringles chip is 35.5cm^2 . This value is between the two values that was found earlier when the surface area was estimate using two methods. From the estimation, I assumed the surface area to be between 27cm^2 and 37cm^2 , therefore the answer from the calculation can be considered reasonable.

4. Conclusion

Through this exploration, I was able to understand the characteristics and how the equation of hyperbolic paraboloid can be used. The investigation on hyperbolic paraboloid made my understanding of complicated shapes better in 3 dimensions. To calculate the surface area of a Pringles chip, a double integral was used. I was able to apply the rules I have learned when I calculated the partial derivatives. Although I attempted to solve the integral of Equation (22), it was unsolvable only with the skills I had learned in the syllabus. However, I understood the steps of calculating double integrals and developed my understanding of calculus beyond the syllabus. At the beginning the double integral seemed easy but it was the most challenging part of the exploration since it was a topic I did not learn in school.

The surface area of a Pringles chip turned out to be 35.5cm^2 which was larger than I initially expected. There was no “official” surface area of a Pringles chip to be found on the internet. However, using two methods, the surface area was estimated and the answer was in the range of two estimations, therefore it could be considered accurate. Still, this could be a limitation of this exploration as it might be less or more than the answer I derived. Another limitation could be the uncertainties in the measurements. A ruler was used to measure the dimensions of a Pringles chip. However, the uncertainties in the ruler should not significantly affect the result as it is a negligible value. In relation to the dimensions, the height above and below the origin was estimated which could also be considered as a limitation. This is because the position of the points would be different which would affect the equation of the hyperbolic paraboloid.

The improvements I could make for future exploration would firstly be to determine more points on a Pringles chip. More points would result in more accurate equation and model of the hyperbolic paraboloid. Also, the thickness of a Pringles chip could be considered if I wanted to find the surface area of a whole Pringles chip. To extend this investigation, the volume of a Pringles chip could be found, involving in triple integration.

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